



Understanding Academic Integrity Beliefs, Motivations and Behaviors Amongst Postgraduate Students: a Comparative Analysis of Public and Private Sector Universities in Punjab, Pakistan

Zaheer Ahmad¹ · Muhammad Shahid Soroya² · Ammara Awais^{3,4}

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Abstract

The increasing reliance on artificial intelligence tools has increased pressure on academic standards, making breaches of academic integrity a clear threat to research quality and scholarship. By using a quantitative survey, this study investigated and compared beliefs, motivations, and behaviors of students from both public and private universities regarding academic dishonesty. The participants were postgraduate students enrolled in six universities, including three public and three private institutions in Punjab, Pakistan. The study found that a lack of instructor guidance on plagiarism, a desire to improve grades, and a fear of failure are significant factors contributing to academic misconduct. Students' behavior, including copying from sources without proper citation and copying from peers, is a common occurrence. Furthermore, the findings suggested that curriculum-embedded integrity education, active supervisory guidance, fostering healthy student-teacher relationships and strong anti-plagiarism policies can build a culture of academic integrity in universities. The research provides evidence-based strategies to promote integrity and reduce misconduct in higher education. It offers a localized perspective on a phenomenon that is rarely explored within the Pakistani higher education landscape.

Keywords Academic integrity · Academic misconduct · Postgraduate students · Plagiarism in universities

✉ Zaheer Ahmad
zaheer.ahmad@uol.edu.pk

¹ Department of Information Management, The University of Lahore, Lahore, Pakistan

² Director General, Institute of Education (PIE), Govt of Pakistan, Islamabad, Pakistan

³ Bayes Business School, City University of London, London, UK

⁴ Lahore Business School (LBS), The University of Lahore, Lahore, Pakistan

Introduction

Academic misconduct poses a significant challenge to the global academic community, as it is considered one of the primary obstacles to innovation and intellectual productivity (Hughes & Eaton, 2022; Khatri, & Karki, 2023). Academic integrity is a fundamental obligation that must be upheld to enhance institutional credibility and reputation (Ali et al., 2023). Trustworthy scholarship is not merely a moral duty; it also serves as a key measure of an institution's reputation and self-branding. Because integrity underpins all of higher education, it shapes the campus culture of honesty, fairness, pedagogical quality, and shared ethical norms (Khamitov, 2023; Singh et al., 2025). Similarly, beliefs in fairness and honesty, as well as exemplary behaviors of peers consistent with professional conduct, play a crucial role in maintaining and upholding the culture of academic integrity (Igbe et al., 2023).

However, the rise of dishonest academic practices poses a significant threat to the credibility of educational institutions worldwide (Chan, 2024; Tindle et al., 2023). The growing concern about academic integrity breaches in higher education institutions (HEIs) has gained considerable attention among researchers, particularly following the proliferation and misuse of AI tools (Ali et al., 2021; Candal-Pedreira et al., 2024; Chen, 2024). The complexity of this issue calls for a thorough investigation into the factors and underlying beliefs and behaviors that influence academic integrity, as well as strategies for fostering a culture of academic honesty within academic institutions.

Previously published literature has established that there are multiple factors that shape the students' behavior to engage in academically dishonest practices, including individual personality traits, organizational culture, associated norms and external pressure (Aslam et al., 2019; Arshad et al., 2021). More recent work also links cheating to sometimes-dominant goals, misuse of privilege, outside influences and a perception that supervisors do not care (Feenstra et al., 2025). Such beliefs and actions do not appear in a vacuum; they differ between public and private universities, where policies, culture, teaching style and the level of support all shape students' ethical compass (Lupton et al., 2000).

Cheating, plagiarism, academic dishonesty, and other forms of academic misconduct have been explored in numerous studies (Chan, 2024; Krou et al., 2021). Various dimensions of academic integrity related to HEIs have also been studied in many countries (e.g., Krienert et al., 2022; Maral, 2024). The study of Zhao et al. (2024) reported that cheating and plagiarism lead to unethical behavior in the workplace. Past research suggests that if a HEI is found to have low academic integrity, stakeholders will lose faith in the degrees awarded to graduates as well as in their competence and moral character (Chan, 2024; Tindle et al., 2023).

In Pakistan, where higher education institutions are rapidly growing, concerns about the academic ethical standards are gaining increased attention from researchers (Waheed, 2024). The Higher Education Commission (HEC), Pakistan, issued an anti-plagiarism policy in 2023 to promote academic integrity among HEIs (HEC, 2023). However, the implementation is inconsistent, and it also relies primarily on punitive measures rather than educational ones (Sivasubramaniam, 2024). Soroya et al. (2020) found that demographic variables, including age and gender, don't affect students' perceptions of plagiarism; how-

ever, the level of education significantly contributes to the awareness of plagiarism among postgraduate students in Pakistan. Research on the motivations behind academic dishonesty among university students in Pakistan is limited (Asgher et al., 2023). Understanding these motivations is essential for developing effective strategies to foster ethical behavior in educational institutions (Kamal, 2023). There is limited research and empirical understanding of beliefs, factors and behavior associated with academic dishonesty, particularly with regard to the public and private sector universities of Pakistan. There is a dire need to investigate this phenomenon to facilitate more effective interventions that promote academic integrity among HEIs in Pakistan.

Academic institutions' reputations depend on their ethical conduct and academic honesty, and upholding these values is their primary goal (Bieliauskaite, 2021; Miron et al., 2021). To give academic institutions a competitive edge, the study provides relevant stakeholders with a clear picture and understanding of the key issues negatively impacting institutional branding. Based on this insight, stakeholders can prepare relevant policies to promote academic honesty in the workplace. Resultantly, this growing academic honesty strengthens the educational landscape in terms of ethical conduct. The research contributes to the development of interventions and initiatives that promote a culture of integrity, improve educational quality, and support the success and learning outcomes of postgraduate students. Additionally, the present study recommends institution-specific strategies to discourage integrity breaches and promote ethical and research integrity among students.

Research Gap

It is a fact that academic misconduct is a severe issue in higher education institutions (Birks et al., 2020; But et al., 2024). Academic integrity in higher education institutions has been studied in many countries (e.g., Balbuena & Lamela, 2015; Maral, 2024) with varying dimensions. Several studies have been conducted to identify various dimensions of academic integrity, like the status of academic integrity in Pakistan (Quraishi & Aziz, 2017; Soroya et al., 2017), perceptions of academic dishonesty among Pakistani students (Arshad et al., 2021; Ellahi et al., 2013; Nazir & Aslam, 2010; Quraishi & Aziz, 2017). The existing literature (i.e., Awasthi, 2019; Haq et al., 2020; MacLeod & Eaton, 2020) has found that multiple contextual factors, including the lack of academic policies, institutional culture, weak student-teacher relationships, and resource inequalities, negatively influence the culture of academic integrity.

Several studies have examined academic misconduct among students in HEIs (e.g., Chan, 2024; Maral, 2024; Soroya et al., 2020). Other studies (e.g., Nazir & Aslam, 2010; Quraishi & Aziz, 2017; Soroya et al., 2020) focused on topics other than the comparative analysis of academically dishonest practices between public and private-sector universities in Punjab, Pakistan. This research gap highlights the need for a thorough investigation of this phenomenon. Additionally, prior studies focused only on general characteristics, attitudes and behavior linked to academic dishonesty. Therefore, this study was designed to provide localized insights into the contextual factors, beliefs, and behaviors that influence breaches of academic integrity.

Study's Objectives

The study has been designed with the following objectives:

1. To identify factors, beliefs and behaviors influencing students to commit academic misconduct in universities.
2. To investigate the level of differences in belief and behavior of postgraduate students related to academic misconduct among Public and Private Sector Universities in Punjab.
3. To propose effective strategies for addressing and reducing academic misconduct among postgraduate students in Punjab.

Literature Review

Academic misconduct encompasses various unethical behaviors, including data falsification, unethical collaboration, plagiarism, and cheating on exams. Such behaviors appear too frequently in universities, disrupting both classroom culture, student performance, and learning outcomes (Ali et al., 2024; Awasthi, 2019). Numerous studies have identified several factors that contribute to academic misconduct among postgraduate students (i.e., Chan, 2024; Haq et al., 2020; Muhammad & Rabbani, 2021). It has been determined that lack of proper understanding, academic pressure, and inconsistent teacher response are the crucial factors (Ali et al., 2024; But et al., 2024). Many learners bend the rules because parents demand top results, stiff competition looms, and the wish to achieve perfect grades is overpowering (Awasthi, 2019). Academic dishonesty is also influenced by the mounting academic stress, challenges in time management, and a lack of academic competencies (Krou et al., 2021). Zhang (2025) found that working stress, publishing in top journals, teachers' pressure, and pressing deadlines are among the key reasons why scholars engage in academically dishonest practices. Saleem (2021) reported that cheating, plagiarism, falsification and failure of the system are the most common factors contributing to academic dishonesty. Students may indulge in plagiarism or other unethical behavior due to a lack of preparation and time constraints.

Several factors, including cultural norms, educational influences, ethical leadership, weak enforcement mechanisms, and a lack of academic support systems, shape students' beliefs about academic dishonesty (Feenstra et al., 2025; Tindle et al., 2023). Zhang (2023) found that both genders engage in misbehavior practices; however, female supervisors tend to be stricter in upholding ethical values compared to their counterparts. The study further recommended that ethical leadership can play a crucial role in shaping students' moral efficacy and promoting an ethical culture within the educational system. Empathy and mindfulness are significant attributes that shape the students' beliefs about academic integrity. Tindle et al. (2023) and Chan (2024) reported that academic misconduct is increasing with the growing use of AI tools by students, especially those who were previously involved in academic misconduct before the proliferation of generative AI.

Juan et al. (2022) found that attitudes and control beliefs influence intentions and justifications for committing academic dishonesty. They further concluded that social norms

within the community, as well as the extent to which students are socialized and connected to these norms, can significantly impact their willingness to commit academic misconduct. Another study conducted by Asgher (2022) found that societal pressure, sense of competition, and social rejection are the determinants of academic dishonesty among students. Kokkinos and Antoniadou (2024) found that psychopathy, dark triad traits and moral disagreement are crucial determinants of the menace of dishonesty. Various behaviors and strategies have been observed in relation to academic integrity breaches among postgraduate students. Commonly reported misconduct behavior includes data fabrications, falsifications, unethical collaborations and exam cheating (Chawla, 2023). Students may be engaged in such behaviors because the perceived benefits outweigh the potential risks.

In a study conducted by Pipere (2022), the author proposed that implementing plagiarism policies properly and holding regular awareness sessions are effective ways to promote academic integrity among university students. Ghimire et al. (2024) found that cheating is associated with the expectation of scholarships and pressure from family members to achieve top grades. Haq et al. (2020) reported three major causes in higher education: mindless memorization, inconsistent grading standards, and peer pressure. They also recommended that schools provide targeted training and mentoring for both staff and students to address the issues. Implementing active teaching practices, academic literacy programs, providing active support services, curriculum-embedded education, and providing necessary tools, along with fostering a healthy teacher-student relationship, can help mitigate the menace of academic dishonesty (Morris, 2023).

Ali et al. (2024) suggested that the enforcement of institutional policies, a culture of ownership and accountability, and active collaboration among stakeholders can help promote academic honesty and raise educational standards. Amin (2024) proposed that conducting regular training, a comprehensive policy framework and student awareness sessions can reduce academic integrity breaches. Well-crafted regulations and policies can foster a culture of integrity on campus over time (Mack & Byanjankar, 2025). Atikuzzaman and Yesmin (2023) and Awais et al. (2025) strongly recommend placing a well-crafted and understandable policy mechanism to promote academic integrity in academic institutions, which can further lead to an increase in the university's overall ranking. Lund et al. (2025) also found that clear policies and proper training can be effective in discouraging academic dishonesty among students. Institutions should, therefore, hold regular workshops that enhance the emotional intelligence of both faculty and research students, as suggested by Tatum and Curtis (2023), thereby reducing the incidence of academic misconduct. In Pakistan, But al. (2024), Ellahi et al. (2013), Haq et al. (2020), and Kamal et al. (2023) also recommended that the enforcement of policies, regular training, and awareness sessions can help reduce academic dishonesty practices among students.

There is a lack of research that compares academic misconduct explicitly between public and private sector universities in Punjab, Pakistan (Zia & Khan, 2023). Understanding the factors, beliefs, and behaviors associated with academic misconduct among postgraduate students is essential for developing effective interventions and preventive measures. Hence, additional work is required to map the unique patterns of misconduct at each type of institution and to design tailored strategies that address the problem in a meaningful way (Saeed, 2023; Zia & Khan, 2023).

Research Design and Method

To meet the study's objectives, the present research employed a quantitative research approach, as it is more suitable when the researcher aims to investigate beliefs, attitudes, perceptions, and behaviors towards a particular phenomenon (Creswell & Clark, 2018). The survey method was employed to gather participants' perceptions about factors, beliefs, and behaviors related to academic misconduct. We obtained permission from the ethical review committee of the university where the research was being conducted. It was clearly stated on the online circulated questionnaire that participation is voluntary, and the data collected for the study will be kept highly confidential. The survey participants' anonymity was ensured.

Population, Sample and Sampling Technique

The study population consisted of all postgraduate students enrolled in the six universities (listed in general categories) in Punjab, Pakistan, comprising three public and three private sector universities. The postgraduate program includes all masters and doctoral students irrespective of their study stages. Those universities were included where academic integrity policies and relevant initiatives are in place to promote integrity values among their students. However, the names of those selected universities have not been disclosed to ensure anonymity. This is also in line with research practices and to protect the integrity of the research context. There are almost 16,800 postgraduate students enrolled in these six universities. The study used Yamane's (1967) formula to draw an appropriate sample size from the population. Some studies (e.g., Ahmad et al., 2023; Saputra & Ariyanto, 2019) employed the same formula for convenience sampling and estimated sample sizes for their respective studies. The formula is given below:

$$n = \frac{N}{1 + N(e)^2}$$

Where; N =Population size, n =Sample size, e =Error of sampling=0.05.

According to the formula, the sample size was 390, with a 95% confidence level and a 5% margin of error. The questionnaire was implemented through the Google Forms platform, consisting of two sections. The first section was used for obtaining informed written consent, while the second section collected demographics and information about all constructs. The questionnaire was disseminated via various institutional social media groups of the concerned universities and the friendship chain technique. A total of 318 students provided their feedback, resulting in an 81% response rate.

Instrument Development

The adapted questionnaire was used as a data collection tool. The scale of factors contributing to committing academic misconduct consists of 12 items, and 10 of them were taken from Stone et al. (2009). Keeping in view the context of the study, two statements, i.e., lack of awareness about institutional plagiarism policy and non-implementation of institutional plagiarism policy, were self-structured. Each factor contributing to academic misconduct was measured on a 5-point Likert scale, ranging from "very unlikely (1)" to "very

likely (5).” The scale of students’ beliefs about academic misconduct was derived from the study by Lupton et al. (2000), and response options ranged from “strongly disagree” (1) to “strongly agree” (5). While measuring academic misconduct behavior, the scale used by Stone et al. (2009), which ranges from “never (1)” to “almost always (5),” was adopted. Following the in-depth literature review, a self-structured list of academic integrity prospects was prepared, and respondents were asked to indicate their preferred approach as a way forward to minimize academic misconduct. The final questionnaire is attached in Appendix A.

Pilot Study

Data was analyzed using IBM SPSS Statistics (version 28). Mean scores, standard deviations, frequencies, and percentages were calculated for descriptive analysis. The instrument was piloted with 40 students having the same characteristics of the selected population. Cronbach’s Alpha values for the scale factors contributing to committing academic misconduct were 0.71, for students’ beliefs about academic misconduct, 0.8, for academic misconduct behavior, and 0.81 for the way forward to minimize academic misconduct. Table 1 displays the details.

Results

The following are the results of the study.

Demographic Information

A total of 318 postgraduate students provided their feedback. During the data collection, it was ensured that the sample was representative of all the selected universities. The demographic information, including gender, university, institution type, and types of students, is presented in Table 2. As seen in Table 2, there were 165 males and 153 females. Out of the total respondents, 146 students were from public sector universities and 172 were from private sector universities who responded to the questionnaire. From the respondents, 282 were full-time students (regular classes from Monday to Friday) and 36 were part-time students (weekend classes).

Factors Contributing to Academic Misconduct

The summary of the statistics, as shown in Table 3, indicates that a lack of instructors’ guidance is the primary factor contributing to academic misconduct. The same findings have

Table 1 Results of pilot study

Scales	Cronbach’s Alpha
Factors contributing to committing academic misconduct	0.70
Students’ beliefs about academic misconduct	0.86
Academic misconduct behavior	0.75
Way forward to minimize academic misconduct	0.81

Table 2 Demographic information

Factor	Description	No
Gender	Male	165
	Female	153
Type of University	Public	146
	Private	172
Types of Students	Full-Time	282
	Part-Time	36

Table 3 Factors contributing to academic misconduct

Factors	Mean	SD	Ranking
Lack of the instructor's guidance	4.3	0.90	1
To increase my grades	4.11	0.98	2
Fear of failure	4.00	0.86	3
Lack of awareness about IPP	3.95	1.03	4
Non-implementation of IPP	3.89	1.00	5
Time pressure	3.90	1.03	6
Laziness	3.86	0.97	7
Peer pressure	3.71	1.01	8
Other students do it	3.51	1.07	9
Extenuating circumstances	3.49	1.04	10
Monetary reward	3.29	1.23	11
To help friend	2.78	1.04	12

Scale Used: 1=Very unlikely
 2=Unlikely, 3=Neutral,
 4=Likely, 5=Very likely *SD =
 Standard Deviation

been endorsed by other studies (e.g., Ali et al., 2024; But et al., 2024). Increasing grades and fear of failure are the second and third major factors that push students to commit academic misconduct. Awasthi (2019) also established that seeking higher grades is the reason for engaging in academic misconduct. A lack of awareness and non-implementation of institutional plagiarism policies are also significant factors contributing to academic misconduct. The same findings have been endorsed by Pipere (2022), Ali et al. (2024) and Amin (2024). Among other ranked factors are time pressure, laziness, peer pressure, other students doing it, extenuating circumstances, monetary reward and helping a friend, which can contribute to academic misconduct in the university. The studies conducted by Feenstra et al. (2025), Tindle et al. (2023), Asgher (2022) and Zhang (2025) also corroborate these findings.

Students' Beliefs about Cheating

Table 4 presents the results of students' beliefs regarding cheating, including mean scores, standard deviations (SD), and rankings of factors based on students' perceptions of cheating. Respondents were provided with nine statements and asked to record their responses. The results indicate that most students believed the instructor's discussion of issues related to cheating could reduce cheating among them. Students also believed that teachers must ensure that instructors make it clear to students that cheating is not acceptable. The studies (Ali et al., 2024; But et al., 2024) also found that regular teacher guidance about the dishonest academic practices can help to curb the menace of plagiarism. This is the second-ranked factor in terms of the student's beliefs about cheating in an academic context. The statement "most students cheat on exams" is the third-ranked factor, indicating that many of their

Table 4 Students' beliefs about cheating

Factors	Mean	SD	Ranking
Instructor's discussion of issues tied to cheating reduces cheating	4.08	0.86	1
Instructors must make sure students do not cheat	4.05	0.94	2
Most students cheat on exams.	3.86	1.04	3
Students are believed to cheat on exams.	3.67	1.03	4
Most students cheat on out-of-class assignments.	3.65	1.04	5
Using an exam from a prior semester is cheating.	3.60	1.05	6
Cheating on one exam is not so bad.	3.51	1.11	7
OK to tell someone in a later section about an exam.	3.39	1.13	8
Giving someone your past exams is cheating	3.30	1.20	9

Scale Used: 1=Strongly Disagree, 2=Disagree, 3=Neutral, 4=Agree, 5=Strongly Agree *SD = Standard Deviation

peers are involved in cheating during exams. This belief is again a contributing factor to committing academic misconduct. The mean score of students believed to cheat on exams is 3.67, ranking 4th, which indicates that students believe their peers are often suspected of cheating during exams. The same results were reported in the studies by Arshad et al. (2021) and Igbe et al. (2023).

There is a belief that cheating on out-of-class assignments is relatively common and it is ranked 5th in the factor. The factor "using an exam from a prior semester is cheating" is ranked 6th, indicating that, on average, students believe this factor constitutes cheating. The standard deviation of 1.05 suggests some diversity in the participants' opinions. The students' beliefs about other factors and their ranking are given in Table 4.

Behavior of Academic Misconduct

The findings are summarized in Table 5, which indicates the prevalence of academic misconduct behavior among postgraduate students of both public and private universities in Punjab. It presents a breakdown of various academic misconduct behaviors, each rated on a scale of 1 to 5, where 1 represents "never" and 5 represents "almost always." Notably, the most prevalent academic misconduct behavior among the students, as indicated by the highest mean score, i.e., 4.24, is "copied a few sentences from a source but not given credit". This behavior shows that most of the students are engaged in this form of academic misconduct behavior. The second-ranked behavior is "copied from another student and turned in as one's own", as indicated by its mean score, i.e., 4.12. It implies that the practice of copying text from a friend's assignment and presenting it as one's own work is widespread among students at the surveyed universities in Punjab. Sozon et al. (2025), through a systematic literature review, concluded that copying without giving credit is common among students. Prashar et al. (2024) and Saeed (2023) have also reported similar results, recommending that the issue can be minimized by providing students with necessary training.

Other behaviors, such as "turned in work done by others", "used unfair methods to learn about a test", and "cheated on a test in any way" also show the higher mean score and indicate that these behaviors are also prevalent among the students, which need to be addressed

Table 5 Behavior of academic misconduct

Factors	Mean	SD	Ranking
Copied a few sentences from a source but not given credit	4.24	0.89	1
Copied from another student and turned it as my own	4.12	0.90	2
Turned in work done by others	4.07	0.96	3
Used unfair methods to learn about a test	4.02	0.91	4
Cheated on test in any way	3.85	1.04	5
Received substantial help on assignment without permission	3.61	1.12	6
Helped someone cheat on a test	3.51	1.07	7
Copied from another student on the test	3.05	1.14	8
Used notes on the test without the instructor's permission	2.99	1.08	9
Collaborated on an assignment that was supposed to be individual work	2.96	1.08	10

Scale Used 1 = Never,
2 = Seldom, 3 = Sometimes,
4 = Often, 5 = Almost always
*SD = Standard Deviation

to discourage academic misconduct. On the other hand, behaviors ranked 8 to 10 show lower mean scores, indicating less frequent engagement in these misconduct activities. These results align with the findings of previous studies (Awdry & Groves, 2023; Chiang et al., 2022), which indicate that exam-related cheating is a widespread issue in academia.

Comparative Analysis between Public and Private Institutions about Beliefs of Cheating

An independent sample t-test was used to compare the mean differences of each belief factor between public and private sector universities. The test details are presented in Table 6. Although a slight mean difference of various factors was observed, the p-values of all the factors, as per the results of Levene's test, were not significant. Students enrolled in public sector universities believed that instructors' discussions of issues related to cheating reduce academic dishonesty, a belief that differs slightly from the opinions gathered from students enrolled in private sector universities. Other differences in beliefs and their ranking are shown in Table 6.

The overall results indicate that there is no statistically significant difference in beliefs about cheating between postgraduate students from public and private sector universities. The results suggest that institutional affiliations (both public and private universities) did not significantly impact beliefs about cheating among postgraduate students. The mean scores for each belief factor in both groups were relatively close, and the standard deviations indicate that the data points were not widely dispersed.

Comparative Analysis of Academic Misconduct Behavior

The independent sample t-test was used to assess the mean differences in various academic misconduct behavior. The data, as shown in Table 7, indicate some interesting results. First, students from public sector universities, on average, reported slightly higher instances of copying a sentence from a source without giving credit and copying from another student and passing it off as their own, when compared to students at private sector universities.

Table 6 Students' beliefs about cheating

Factors	N	Institution Types	Mean	SD	Ranking
Instructors' discussion of issues tied to cheating reduces cheating	146 172	Public Private	4.14 4.04	0.88 0.84	1
Instructors must make sure students do not cheat	146 172	Public Private	4.13 3.98	0.82 0.80	2
Most students cheat on exams.	146 172	Public Private	3.90 3.82	1.02 1.05	3
Students are believed to cheat on exams.	146 172	Public Private	3.68 3.66	0.98 1.06	4
Most students cheat on out-of-class assignments.	146 172	Public Private	3.64 3.65	1.08 1.01	5
Using an exam from a prior semester is cheating.	146 172	Public Private	3.64 3.58	1.04 1.06	6
Cheating on one exam is not so bad.	146 172	Public Private	3.46 3.55	1.14 1.07	7
OK to tell someone in a later section about an exam.	146 172	Public Private	3.32 3.45	1.16 1.09	8
Giving someone your past exams is cheating	146 172	Public Private	3.26 3.33	1.20 1.20	9

Scale Used: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree *SD = Standard Deviation

Additionally, there was a notable difference in the use of unfair methods to learn about the test. Students at public sector universities reported a slightly higher frequency of this behavior compared to their counterparts in private universities. Students of public sector universities reported a higher frequency of receiving help in preparing an assignment without the permission of the original contributor as compared to the students of private sector universities.

However, there were no statistically significant differences in the mean values of all other behaviors of academic misconduct between postgraduate students in public and private sector universities. Examining the overall pattern indicates that there are relatively similar levels of misconduct behavior among the postgraduate students of both public and private universities in Punjab.

Way Forward to Minimize Academic Misconduct

Table 8 outlines various proposed strategies to combat academic misconduct and each factor is assessed by respondents on a scale of 1 to 5, where 1 represents strongly disagree and 5 represents strongly agree. The ranking of the factor highlights the order of importance, as assessed by the students, to combat academic misconduct. Notably, the factor that ranked top is “educate academic integrity through course-based education” with a mean score of 4.28. The results are aligned with the findings of Bens (2022) and Morris (2023) who emphasized the embedding of academic integrity content within formal educational courses to mitigate dishonest practices in academia. It implies that the majority of postgraduate stu-

Table 7 Behavior of academic misconduct

Factors	N	Institution Types	Mean	SD
Copied a few sentences from a source but not given credit	146	Public	4.28	0.89
	172	Private	4.23	0.89
Copied from another student and turned in as own	146	Public	4.24	0.89
	172	Private	4.01	0.89
Turned in work done by others	146	Public	4.11	0.91
	172	Private	4.03	1.00
Used unfair methods to learn about a test	146	Public	4.04	0.90
	172	Private	3.99	0.93
Cheated on test in any way	146	Public	3.88	1.00
	172	Private	3.81	1.07
Received substantial help on assignment without permission	146	Public	3.54	1.18
	172	Private	3.67	1.06
Helped someone cheat on a test	146	Public	3.44	1.09
	172	Private	3.55	1.06
Copied from another student on the test	146	Public	3.03	1.16
	172	Private	3.06	1.24
Used notes on the test without the instructor's permission	146	Public	2.91	1.10
	172	Private	3.05	1.06
Collaborated on assignment that was supposed to be individual work	146	Public	2.88	1.09
	172	Private	3.02	1.06

Scale Used: 1=Never, 2=Seldom, 3=Sometimes, 4=Often, 5=Almost always
 *SD = Standard Deviation

Table 8 Way forward to minimize academic misconduct

Factors	Mean	SD	Ranking
Educate academic integrity through course-based education	4.28	0.91	1
Supervisory guidance on academic integrity	4.23	2.00	2
Effective faculty-student relationships	4.18	1.97	3
Regular training and awareness of academic integrity	4.14	1.96	4
Institutional library's role in promoting academic integrity	4.08	0.95	5
Implementation of IPP	3.98	0.90	6
Clear policy and guidelines on academic integrity	3.97	1.96	7
Availability and access to plagiarism detection tools	3.95	1.03	8
Provide support and educational resources on academic integrity	3.90	0.96	9
Disciplinary actions against academic misconduct	3.59	1.10	10

Scale Used: 1=Strongly Disagree, 2=Disagree, 3=Neutral, 4=Agree, 5=Strongly Agree *SD = Standard Deviation

dents strongly support the idea of incorporating academic integrity education directly into their academic courses.

The “supervisory guidance on academic integrity” ranked second with a mean score of 4.23. It indicates that the students value the role of the teacher and advisor in guiding them to avoid academic misconduct. “Effective faculty-student relationships” and “regular training and awareness of academic integrity” also received high mean scores, ranking third

and fourth, respectively. These results validate the results of MacLeod and Eaton (2020), and Prashar et al. (2024). The results of these studies further recommend that a healthy student-teacher relationship and regular training sessions can be effective in maintaining academic integrity. Participants also prioritized other strategies to promote academic integrity, including the role of the institutional library, effective enforcement of the institutional plagiarism policy, and clear policies and guidelines on academic integrity. These findings suggest that such strategies are also vital in reducing academic misconduct among students. These results are consistent with many previous studies (e.g., Ali et al., 2024; Atikuzzaman & Yesmin, 2023; Amin, 2024; Lund et al., 2025; Mack & Byanjankar, 2025).

More importantly, “disciplinary actions against academic misconduct” received a relatively lower mean score, ranking tenth, indicating that this factor is recognized as a strategy but may not be a primary focus for students minimizing academic misconduct in the university. The result aligns with the findings of Zhao et al. (2024), who reported that subtle reminders and low levels of punitive measures can be more effective instead of taking strict disciplinary actions.

The study further suggests that punitive measures may not be the most effective strategy to reduce academic dishonesty. In contrast, Ilahi (2024), emphasized the need for disciplinary actions to mitigate academic dishonesty practices. Sivasubramaniam (2024) and HEC (2023) also emphasized the punitive measures to discourage dishonesty in educational settings. Morris (2023) also suggested that strict punishment is the most effective measure to prevent academic dishonesty among university students. This diverse opinion on the same strategy may be due to cultural norms prevailing in that specific society.

Discussion

The study identified factors that contribute to cheating, such as a lack of instructor guidance and a desire to improve grades, which align with previous research (Coghlan et al., 2021). Based on these findings, the study suggests that the university administration provide clear guidance and emphasize the importance of adhering to ethical principles to prevent academic misconduct. Prior studies (Ali et al., 2024; But et al., 2024) also indicated that policy awareness and active supervisory support can significantly enhance students’ understanding and attitudes toward maintaining integrity in research. Additionally, the fear of failure and the absence of institutional policies to address plagiarism are factors that contribute to academic misconduct (Chan, 2024; Ifeagwazi et al., 2019). These issues can be addressed effectively through strong student mentoring.

When examining the behavior of academic misconduct among postgraduate students, actions such as copying a few sentences from a source without proper credit (Sozon et al., 2025; Prashar et al., 2024), copying from another student and submitting it as one’s own (Saeed, 2023), using unfair methods to learn about a test, and receiving significant help on assignments without permission are common. The lack of awareness about correct citation practices often stems from insufficient training, which can be addressed through targeted initiatives. This suggests that institutions should adopt a proactive approach to mentoring and providing regular training to students, thereby minimizing academic misconduct.

It was interesting to observe that there was no significant difference between the beliefs and behaviors of postgraduate students regarding academic misconduct in public and pri-

vate universities. However, the overall patterns did not vary significantly. Findings suggest that postgraduate students from both types of institutions perceive and act similarly toward academic misconduct, indicating that these patterns reflect a shared trend among Pakistani educational institutions rather than being specific to any one institution. The findings support the adoption of a standardized approach, including the creation of uniform policies, guidelines, and training programs, to promote academic integrity across all universities, whether private or public (Ali et al., 2024; But et al., 2024). Furthermore, collaborative efforts and regulatory bodies involving both public and private institutions can play a crucial role in reducing academic misconduct practices in universities (Ali et al., 2024).

The study recommends that course-based education can effectively promote academic integrity among students, aligning with Rahimian (2023). It highlights the importance of including academic integrity in the curriculum to address this issue in higher education. Similar findings are reported by Bens (2022) and Morris (2023). The university administration should involve all relevant stakeholders, including faculty, students, administrative staff, librarians, and IT support personnel, and develop course-based instructions and training programs to raise students' awareness. Collaborative efforts can lead to better results in fostering a shared culture of integrity across campus (Rasul et al., 2024). As a central educational hub, libraries should play a vital role in conducting awareness and training programs for all university students. Academic libraries should develop information and AI literacy modules to enhance capacity among all relevant stakeholders in academic integrity and credibility.

Additionally, supervisory guidance on academic integrity (Cutri et al., 2021; Coghlan et al., 2021), student-teacher relationships (Cutri et al., 2021; MacLeod & Eaton, 2020), ongoing training, and awareness campaigns (Morris, 2023; Rahimian, 2023; Younis & Gishen, 2019) are considered the most effective ways to prevent academic misconduct. Yusuf et al. (2024) also found that students often misuse technology, including cheating, especially when strong student-teacher relationships are lacking, highlighting the need to foster solid connections.

Creating a strong culture of academic honesty in higher education institutions (HEIs) involves establishing clear standards, well-defined norms, and consistent enforcement to encourage values of academic integrity. Previous studies (e.g., Ali et al., 2024; Artyukhov, 2024) support these findings. The results also identified the institutional library as a key factor in promoting academic integrity values. Other studies have also recognized the library as an essential provider of plagiarism detection tools and advisory services to all stakeholders. Libraries also play a crucial role in providing literacy services and awareness programs, emphasizing their active role in combating plagiarism in academic settings (Gunton, 2022). This finding is supported by Mansoor et al. (2023). This study established that the library's potential can be effectively utilized in promoting academic integrity, with strong support from relevant institutions. Prior studies (e.g., Sivasubramaniam, 2024) and initiatives undertaken by the HEC (2023) in Pakistan have primarily relied on punitive measures to discourage academic dishonesty; however, this study suggests that it is the least prioritized recommendation among the participants.

It implies that rather than relying more on punitive measures, institutions need to prioritise incorporating course-based instruction, proper supervisory guidance mechanisms, regular training, and mentoring of students, which can better help promote academic honesty

among academic institutions. It can further be argued that over-reliance on punishment can lead to resistance rather than long-term behavioral change.

The study's important limitations include the data collected from only the three private and public universities in Punjab, and insights from the remaining university students are not reported. As the data were collected from a specific region of Punjab and post-graduate students only, this limits the broader applicability of the findings. The population doesn't differentiate between the opinions of STEM and non-STEM postgraduate students despite the fact that ethical practices may vary between of them. Furthermore, the data were collected using a cross-sectional approach, and the behavior and beliefs change over time, so generalizability is limited.

Conclusion and Recommendations

The study's findings provide valuable insights about students' perceptions, beliefs, and behaviors regarding academic misconduct in university settings. The study also suggested a possible way forward to enhance the culture of academic integrity while discouraging academic dishonesty.

The study identified determinants that contribute to students engaging in dishonest academic practices. These factors include a lack of proper guidance from the supervisor, a desire to achieve good grades, fear of failure, inadequate institutional policies, peer pressure, and time constraints. The institutions need to address these factors to minimize breaches of academic integrity among students. Secondly, Students think that appropriate guidance and mentoring from teachers can help reduce the problem of academic dishonesty. In addition, university students believe that cheating during exams is a major problem that requires appropriate counselling and awareness regarding the consequences of such behaviour.

Thirdly, the study found that the most frequent misbehavior is to copy someone else's idea and portray it as their own. This prevalent student behaviour highlights the importance and need of regular training on citation management. Capacity-building initiatives focused on citation management can empower students to master the art of acknowledging and citing sources within their text. Fourthly, the survey asked questions about attitudes and behaviors concerning dishonesty for both public and private universities. These results showed that the attitudes and behaviors concerning academic dishonesty for both private and public postgraduate students differed slightly. Despite that, the difference was not statistically significant.

Lastly, the study offered effective strategies to help reduce dishonest practices among students. Suggestions include course-integrated teaching, active teacher guidance on the values of academic integrity, and strong student-teacher relationships. Educational institutions should develop and implement relevant policies that promote a culture of integrity among students. Surprisingly, the study did not find any difference in academic misconduct behaviors between public and private educational institutions in Pakistan. The results indicate that students' beliefs and behaviors regarding academic misconduct are similar in both types of institutions. All universities, regardless of whether they are private or public, should recognize that it is their shared responsibility to create a culture of academic integrity on campus, which will further enhance the institution's reputation.

In conclusion, these findings provide valuable insights and a thorough understanding of the factors that lead to academic misconduct, students' perceptions of cheating, common

misconduct behaviors, and practical strategies for addressing academic misconduct among postgraduate students. The results serve as a guide for universities to develop targeted interventions and relevant institutional policies that promote academic integrity and effectively prevent dishonesty.

Appendix A: Research Questionnaire

Section A: Factors Contributing to Academic Misconduct

Q No. 1. Please rate your level of agreement with the following statements.

No	Statements	Very unlikely (1)	Unlikely (2)	Neutral (3)	Likely (4)	Very likely (5)
1.1	To help a friend					
1.2	Time pressure					
1.3	Extenuating circumstances					
1.4	Peer pressure					
1.5	To increase my grade					
1.6	Other students do it					
1.7	Monetary reward					
1.8	Lack of instructor's guidance					
1.9	Fear of failure					
1.10	Laziness					
1.11	Lack of awareness about institutional plagiarism policy (self-structured)					
1.12	Non-implementation of institutional Plagiarism policy (self-structured)					

Section B: Students' Beliefs about Cheating

Q No. 2. Please rate your level of agreement with the following statements.

No.	Statements	Strongly Disagree (1)	Disagree (2)	Neutral (3)	Agree (4)	Strongly Agree (5)
2.1	Students believed to cheat on exams.					
2.2	Most students cheat on exams.					
2.3	Most students cheat on out-of-class assignments.					
2.4	Cheating on one exam is not so bad.					
2.5	OK to tell someone in a later section about an exam.					
2.6	Giving someone your past exams is cheating.					
2.7	Using an exam from a prior semester is cheating.					
2.8	Instructors must make sure students do not cheat.					
2.9	Instructor's discussing of issues tied to cheating reduces the amount of cheating					

Section C: Behavior of Academic Misconduct

Q. No. 3. Please rate your level of agreement with the following statements.

No	Statements	Never (1)	Sel- dom (2)	Some- times (3)	Often (4)	Al- most always (5)
3.1	Copied a few sentences from a source but not given credit					
3.2	Copied from another student and turned in as own					
3.3	Helped someone cheat on a test					
3.4	Collaborated on an assignment that was supposed to be individual work					
3.5	Turned in work done by others					
3.6	Copied from another student on the test					
3.7	Used notes on the test without the instructor's permission					
3.8	Received substantial help on assignment without permission					
3.9	Cheated on test in any way					
3.10	Used unfair methods to learn about a test					

Section D: Way Forward to Minimize Academic Misconduct

Q. No. 4. Please rate your level of agreement with the following statements.

No.	Statements	Strong- ly Dis- agree (1)	Dis- agree (2)	Neu- tral (3)	Agree(4)	Strong- ly Agree (5)
6.1	Educate academic integrity through course-based education.					
6.2	Disciplinary actions against academic misconduct					
6.3	Clear policy and guidelines on academic integrity					
6.4	Availability and access to plagiarism detection tools					
6.5	Supervisory guidance on academic integrity					
6.6	Institutional library's role in promoting the academic integrity					
6.7	Regular training and awareness of academic integrity					
6.8	Implementation of institutional plagiarism policy					
6.9	Provide support and educational resources on academic integrity					
6.10	Effective faculty-student relationships					
6.11	Any other, please mention.					

Section E: Demographic Information

Gender: ☐ Male ☐ Female

Your University Name: _____

Institution: ☐ Public ☐ Private

Type of students: ☐ Full-Time ☐ Part-Time

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Declarations

Research Involving Human Participants The researchers obtained permission from the university's ethical review committee, where the research was being conducted (Identifier: UOL/IREB/25/10.1/0001). As this study involved human participants, all procedures were conducted in accordance with the ethical standards of the participating institutions.

Informed Consent Before participating, study participants were informed (through a statement written above the questionnaire) about the procedures, benefits, and other aspects. The questionnaire was administered online, and each participant documented their consent before providing feedback. Participation was entirely voluntary.

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References

- Ahmad, Z., Soroya, S. H., & Mahmood, K. (2023). Bonding and bridging social capital as predictors of information sharing intention and behavior among Pakistani Facebook users. *Information Development*, 41(1), 155–170. <https://doi.org/10.1177/02666669231200458>
- Ali, I., Sultan, P., & Aboelmaged, M. (2021). A bibliometric analysis of academic misconduct research in higher education: Current status and future research opportunities. *Accountability in Research*, 28(6), 372–393. <https://doi.org/10.1080/08989621.2020.1836620>
- Ali, M. A., Tariq, S. M., & Khalid, U. (2024). Academic integrity in higher education institutions (HEIs) of Pakistan: Problems and recommendations. *International Journal of Social Science Archives (IJSSA)*, 7(2), 638–648. Retrieved July 06, 2025, from <https://ijssa.com/index.php/ijssa/article/view/339>
- Ali, M. V., Fazal, B., & Tariq, J. (2023). Morality and academic dishonesty among university students: Testing extended theory of planned behavior. *Contemporary Issues in Social Sciences and Management Practices*, 2(4), 20–29. <https://doi.org/10.61503/cissmp.v2i4.81>
- Amin, K. (2024). The role of university management in reducing student's academic dishonesty in Punjab. *Harf-o-Sukhan*, 8(3), 26–36. Retrieved August 06, 2025, from <https://www.harf-o-sukhan.com/index.php/Harf-o-sukhan/article/view/1559/1494>

- Arshad, I., Zahid, H., Umer, S., Khan, S. Y., Sarki, I. H., & Yaseen, M. N. (2021). Academic dishonesty among higher education students in Pakistan. *Elementary Education Online*, 20(5), 5334–5345. Retrieved July 7, 2025 from <https://ilkogretim-online.org/index.php/pub/article/view/4447/4322>
- Artyukhov, A. (2024). University system for ensuring academic integrity: Structure and role of stakeholders. *Baltic Journal of Legal and Social Sciences*, 1, 261–267. <https://doi.org/10.30525/2592-8813-2024-1-32>
- Asgher, S. (2022). Social factors behind academic dishonesty as perceived by the postgraduate students. *International Journal of Social Studies*, 2(2), 49–53. <https://doi.org/10.55627/ijss.02.2.0262>
- Asgher, S., Khan, F. U., Murtaza, G., Abid, S., Shabbir, M., Ashraf, S., & Munawar, B. (2023). Factors leading students to academic dishonesty on the university level in Pakistan. *Journal Of Education And Social Studies*, 4(3), 836–843. <https://doi.org/10.52223/jess.2023.4344>
- Aslam, F., Mahboob, U., & Zahra, Q. (2019). Medical students' perception and involvement in academic dishonesty: A cross-sectional study in Lahore, Pakistan. *Health Professions Educator Journal*, 2(2), 18–26. Retrieved August 23, 2025, from <https://journals.uol.edu.pk/medicaleducator/article/view/19/118>
- Atikuzzaman, M., & Yesmin, S. (2023). Authority concerns regarding research students' academic dishonesty: A case study for promoting academic integrity in a public university in Bangladesh. *Journal of Academic Ethics*, 21(4), 591–607. <https://doi.org/10.1007/s10805-023-09474-8>
- Awais, A., Ahmad, Z., & Soroya, M. S. (2025). Role of policies in preserving research and academic integrity. In A. K. Abu-Shaheen, M. A. Hamza, & S. Marar (Eds.), *Introduction to research ethics and academic integrity* (pp. 99–108). Springer. https://doi.org/10.1007/978-981-96-9678-9_9
- Awasthi, S. (2019). Plagiarism and academic misconduct A systematic review. *DESIDOC Journal of Library & Information Technology*, 39(2), 94–100. <https://doi.org/10.14429/djlit.39.2.13622>
- Awdry, R., & Groves, A. (2023). Why they do and why they don't: A combined criminological approach to understanding assignment outsourcing in higher education. *International Journal for Educational Integrity*, 19(1), Article 7. <https://doi.org/10.1007/s40979-023-00126-3>
- Balbuena, S. E., & Lamela, R. A. (2015). Prevalence, Motives, and Views of Academic Dishonesty in Higher Education. *Online Submission*, 3(2), 69–75. Retrieved August 23, 2025, from <https://files.eric.ed.gov/fulltext/ED575015.pdf>
- Bens, S. (2022). Helping students resolve the ambiguous expectations of academic integrity. In S. E. Eaton, & J. Christensen Hughes (Eds.), *Academic integrity in Canada. Ethics and integrity in educational contexts* (Vol. 1). Springer. https://doi.org/10.1007/978-3-030-83255-1_19
- Bieliauskaitė, J. (2021). Solidarity in academia and its relationship to academic integrity. *Journal of Academic Ethics*, 19(3), 309–322. <https://doi.org/10.1007/s10805-021-09420-6>
- Birks, M., Mills, J., Allen, S., & Tee, S. (2020). Managing the mutations: Academic misconduct in Australia, new Zealand and the UK. *International Journal for Educational Integrity*, 16, 1–15. <https://doi.org/10.1007/s40979-020-00055-5>
- But, A. R., Ahmad, Z., Ali, R. F., & Iqbal, Q. (2024). Academic integrity: A five-decade journey, from cheating to a lack of Understanding of academic dishonesty. *Journal of Excellence in Social Sciences*, 3(3), 211–231. <https://doi.org/10.69565/jess.v3i3.403>
- Candal-Pedreira, C., Ross, J. S., Marušić, A., & Ruano-Ravina, A. (2024). Research misconduct as a challenge for academic institutions and scientific journals. *Journal of Epidemiology and Community Health*, 78(1), 61–64. <https://doi.org/10.1136/jech-2023-220554>
- Chan, C. K. Y. (2024). Students' perceptions of 'AI-giarism': Investigating changes in understandings of academic misconduct. *Education And Information Technologies*. <https://doi.org/10.1007/s10639-024-13151-7>
- Chawla, D. S. (2023). Survey finds cheating is common in graduate school. *Physics*, 16, 90. Retrieved July 12, 2025, from <https://physics.aps.org/articles/v16/90>
- Chen, Z., Chen, C., Yang, G., He, X., Chi, X., Zeng, Z., & Chen, X. (2024). Research integrity in the era of artificial intelligence: Challenges and responses. *Medicine*, 103(27), Article e38811. <https://doi.org/10.1097/MD.00000000000038811>
- Chiang, F. K., Zhu, D., & Yu, W. (2022). A systematic review of academic dishonesty in online learning environments. *Journal of Computer Assisted Learning*, 38(4), 907–928. <https://doi.org/10.1111/jcal.12656>
- Christensen Hughes, J., & Eaton, S. E. (2022). *Academic Misconduct in Higher Education: Beyond Student Cheating* (pp. 81–102). Springer. https://doi.org/10.1007/978-3-030-83255-1_4
- Coghlan, S., Miller, T., & Paterson, J. (2021). Good proctor or big brother? Ethics of online exam supervision technologies. *Philosophy & Technology*, 34(4), 1581–1606. <https://doi.org/10.1007/s13347-021-00476-1>
- Creswell, J. W., & Clark, V. (2018). *Designing and conducting mixed methods research* (3rd ed.). SAGE.

- Cutri, J., Abraham, A., Karlina, Y., Patel, S. V., Moharami, M., Zeng, S., & Pretorius, L. (2021). Academic integrity at doctoral level: The influence of The imposter phenomenon and cultural differences on academic writing. *International Journal for Educational Integrity*, 17(1), 1–16. <https://doi.org/10.1007/s40979-021-00074-w>
- Ellahi, A., Mushtaq, R., & Khan, M. B. (2013). Multi campus investigation of academic dishonesty in higher education of Pakistan. *International Journal of Educational Management*, 27(6), 647–666. <https://doi.org/10.1108/IJEM-03-2012-0039>
- Feenstra, R. A., García, C., C., & Gómez Nicolau, E. (2025). Perception of research misconduct in a Spanish university. *Journal of Academic Ethics*, 23(1), 89–112. <https://doi.org/10.1007/s10805-024-09526-7>
- Ghimire, S. N., Bhattarai, U., & Baral, R. K. (2024). Academic dishonesty within higher education in Nepal: An examination of students' exam cheating. *Journal of Academic Ethics*, 22(2), 303–322. <https://doi.org/10.1007/s10805-023-09486-4>
- Gunton, L. (2022). Librarians as drivers of academic integrity for student success at university. *Journal of the Australian Library and Information Association*, 71(2), 156–170. <https://doi.org/10.1080/24750158.2022.2063101>
- Haq, A. U., Mahmood, S., Shabbir, M., & Batool, Z. (2020). Assessing the academic integrity among university students in Pakistan. *Journal of Business and Social Review in Emerging Economies*, 6(3), 1025–1032. <https://doi.org/10.26710/jbsee.v6i3.1359>
- HEC (Higher Education Commission) Islamabad (2023). Anti-plagiarism policy (V2). In <https://www.hec.gov.pk>. Retrieved September 12, 2025 from <https://www.hec.gov.pk>
- Ifeagwazi, C. M., Chukwuorji, J. C., Egbodo, S. O., & Nwoke, M. B. (2019). Peer pressure, fear of failure and examination cheating behavior in the university: Does gender make the difference? *Cognition Brain Behavior: An Interdisciplinary Journal*, 23(1), 43–62. <https://doi.org/10.24193/cbb.2019.23.03>
- Igbe, F. O., Ethe, N., & Ossai, M. C. (2023). Predictors of examination integrity among secondary school students: Framework for proactive actions against examination malpractice. *Education Quarterly Reviews*, 6(3), 250–259. <https://doi.org/10.31014/AIOR.1993.06.03.779>
- Ilahi, F., Manzoor, T., & Elahi, I. (2024). Enhancing assessment integrity: A critical analysis of transparency and fairness in marking process at University of Sargodha. *Journal of Education and Social Studies*, 5(2), 489–501. <https://doi.org/10.52223/jess.2024.5229>
- Juan, L. X., Tao, W. Y., Veloo, P. K., & Supramaniam, M. (2022). Using extended TPB models to predict dishonest academic behaviors of undergraduates in a Chinese public university. *Sage Open*, 12(4), 1–18. <https://doi.org/10.1177/21582440221140391>
- Kamal, M., Parveen, Z., & Mushtaq, T. (2023). The prevalence of academic dishonesty among university students: An action research study in the field of education. *Pakistan Journal of Educational Research*, 6(3). <https://doi.org/10.52337/pjer.v6i3.906>
- Khamitov, N. (2023). Academic integrity as a challenge, requirement and will: Contexts of philosophical anthropology, ethics and philosophy of education. *Philosophy of Education Philosophy of Education*, 29(2), 27–47. <https://doi.org/10.31874/2309-1606-2023-29-2-2>
- Khatrri, B. B., & Karki, P. D. (2023). Artificial intelligence (AI) in higher education: Growing academic integrity and ethical concerns. *Nepalese Journal of Development and Rural Studies*, 20(01), 1–7. <https://doi.org/10.3126/njdrs.v20i01.64134>
- Kokkinos, C. M., & Antoniadou, N. (2024). Understanding academic dishonesty in university settings: The interplay of dark triad traits and moral disengagement. *The Journal of Genetic Psychology*, 185(5), 309–322. <https://doi.org/10.1080/00221325.2023.2297850>
- Krienert, J. L., Walsh, J. A., & Cannon, K. D. (2022). Changes in the tradecraft of cheating: Technological advances in academic dishonesty. *College Teaching*, 70(3), 309–318. <https://doi.org/10.1080/87567555.2021.1940813>
- Krou, M. R., Fong, C. J., & Hoff, M. A. (2021). Achievement motivation and academic dishonesty: A meta-analytic investigation. *Educational Psychology Review*, 33, 427–458. <https://doi.org/10.1007/s10648-020-09557-7>
- Lund, B. D., Lee, T. H., Mannuru, N. R., & Arutla, N. (2025). AI and academic integrity: Exploring student perceptions and implications for higher education. *Journal of Academic Ethics*. <https://doi.org/10.1007/s10805-025-09613-3>
- Lupton, R. A., Chapman, K. J., & Weiss, J. E. (2000). International perspective: A cross-national exploration of business students' attitudes, perceptions, and tendencies toward academic dishonesty. *Journal of Education for Business*, 75(4), 231–235. <https://doi.org/10.1080/08832320009599020>
- Mack, L., & Byanjankar, A. (2025). Understanding policy responses to generative AI in private higher education: Insights from Japan and the united States. *Journal of Academic Ethics*, 23(2), 2357–2373. <https://doi.org/10.1007/s10805-025-09655-7>
- MacLeod, P. D., & Eaton, S. E. (2020). The paradox of faculty attitudes toward student violations of academic integrity. *Journal of Academic Ethics*, 18(4), 347–362. <https://doi.org/10.1007/s10805-020-09363-4>

- Mansoor, F., Ameen, K., & Arshad, A. (2023). Antiplagiarism practices of university libraries in Pakistan: Perceptions of library directors. *New Review of Academic Librarianship*, 29(3), 304–325. <https://doi.org/10.1080/13614533.2023.2196261>
- Maral, M. (2024). A bibliometric analysis on academic integrity. *Journal of Academic Ethics*, 22(1), 665–687. <https://doi.org/10.1007/s10805-024-09519-6>
- Miron, J., Eaton, S. E., McBreaity, L., & Baig, H. (2021). Academic integrity education across the Canadian higher education landscape. *Journal of Academic Ethics*, 19(4), 441–454. <https://doi.org/10.1007/s10805-021-09412-6>
- Morris, E. J. (2023). Integrating academic integrity: An educational approach. In S. E. Eaton (Ed.), *Handbook of academic integrity*. Springer. https://doi.org/10.1007/978-981-287-079-7_96-1
- Muhammad, H. J., & Rabbani, A. (2021). Academic integrity, cheating and academic performance among university students GC University Lahore, Pakistan. *Pakistan Journal of Social and Clinical Psychology*, 19(1), 43–49. Retrieved July 12, 2025, from <https://gcu.edu.pk/pages/gcupress/pjscp/volumes/2021-1/6-Academic-Integrity-Cheating-and-Academic-Performance.pdf>
- Nazir, M. S., & Aslam, M. S. (2010). Academic dishonesty and perceptions of Pakistani students. *International Journal of Educational Management*, 24(7), 655–668. <https://doi.org/10.1108/09513541011080020>
- Pipere, A. (2022). Meta-analyses on teachers' and students' views on academic integrity and a way forward. In *Proceedings of Economics and Finance Conferences* (No. 13415695). International Institute of Social and Economic Sciences. Retrieved July 25, 2025, from <https://ideas.repec.org/p/sek/iefpro/13415695.html>
- Prashar, A., Gupta, P., & Dwivedi, Y. K. (2024). Plagiarism awareness efforts, students' ethical judgment and behaviors: A longitudinal experiment study on ethical nuances of plagiarism in higher education. *Studies in Higher Education*, 49(6), 929–955. <https://doi.org/10.1080/03075079.2023.2253835>
- Quraishi, U., & Aziz, F. (2017). Academic Dishonesty at the Higher Education Level in Punjab, Pakistan. *Journal of Research & Reflections in Education (JRRE)*, 11(1). Retrieved 6 July 2025, from <https://ue.edu.pk/jrre/articles/1101007.pdf>
- Rasul, T., Nair, S., Kalendra, D., Balaji, M. S., de Oliveira Santini, F., Ladeira, W. J., & Hossain, M. U. (2024). Enhancing academic integrity among students in GenAI era: A holistic framework. *The International Journal of Management Education*, 22(3), Article 101041. <https://doi.org/10.1016/j.ijme.2024.101041>
- Saeed, R. (2023). Academic honesty in undergraduate students in Pakistan. *Dinkum Journal of Medical Innovations*, 2(03), 91–96. Retrieved 10 July 2025, from <https://www.dinkumpublishers.com/djmi/d-0106/>
- Saleem, M., Akram, G., Durrani, A. K., & Siddique, A. R. (2021). Unriddling academic dishonesty through the lens of teacher and student perceptions. *NUST Journal of Social Sciences and Humanities*, 7(1), 63–77. <https://doi.org/10.51732/njssh.v7i1.67>
- Saputra, A., & Ariyanto, E. (2019). The effect of leader member exchange, job satisfaction and motivation on educational personnels' organizational commitment of Jakarta Mercu Buana University. *International Review of Management and Marketing* 9(6): 58–66. Retrieved July 5, 2025 from <https://www.econjournals.com/index.php/irmm/article/view/8650>
- Singh, J. P., Mishra, N., & Mishra, K. K. (2025). Promoting academic integrity: Strategies and challenges in higher education. *Higher Education and Quality Assurance Practices* (pp. 191–224). IGI Global Scientific Publishing.
- Sivasubramaniam, S. D. (2024). Academic integrity in South Asia: Focus on India, Pakistan, and Sri Lanka. *Second Handbook of Academic Integrity* (pp. 77–93). Springer. https://doi.org/10.1007/978-3-031-54144-5_88
- Soroya, D. M. S., Hashmi, D. M. A., & Soroya, S. H. (2020). Academic integrity: Effects of demographic variables on students' conduct. *South Asian studies*, 31(2), 423–438. Retrieved July 09, 2025. <https://sasj.pu.edu.pk/9/article/view/1043>
- Soroya, M. S., Hashmi, M. A., & Soroya, S. H. (2017). Status of academic integrity in Pakistan: An overview. In *plagiarism across Europe and beyond 2017: Conference Proceedings* (pp. 254–262). Brno, Czech Republic: Mendel University.
- Sozon, M., Pok, W. F., Sia, B. C., & Alkharabsheh, O. H. M. (2025). Cheating and plagiarism in higher education: A systematic literature review from a global perspective, 2016–2024. *Journal of Applied Research in Higher Education*, 17(5), 1728–1742. <https://doi.org/10.1108/JARHE-12-2023-0558>
- Stone, T. H., Jawahar, I. M., & Kisamore, J. L. (2009). Using the theory of planned behavior and cheating justifications to predict academic misconduct. *Career Development International*, 14(3), 221–241. <https://doi.org/10.1108/13620430910966415>
- Tatum, H. E., & Curtis, G. J. (2023). Capitalising on emotions and emotional regulation: Five strategies to improve academic integrity. In G. J. Curtis (Ed.), *Academic Integrity in the Social Sciences* (Vol. 6, pp. 69–86). Springer International Publishing AG. https://doi.org/10.1007/978-3-031-43292-7_5

- Tindle, R., Pozzebon, K., Willis, R., & Moustafa, A. A. (2023). Academic misconduct and generative artificial intelligence: University students' intentions, usage, and perceptions. *PsyArXiv Preprints*, 13. <https://doi.org/10.31234/osf.io/hwkgu>
- Waheed, U. (2024). Paper mills and intellectual prostitution in Pakistani research: A challenge to scientific integrity. *Journal of Islamabad Medical & Dental College*, 13(Suppl.), 495–498. [https://doi.org/10.35787/jimdc.v13i\(Suppl.\).1320](https://doi.org/10.35787/jimdc.v13i(Suppl.).1320).
- Yamane, T. (1967). *Elementary sampling theory*. Prentice-Hall.
- Younis, J., & Gishen, F. (2019). Practical tips for teaching academic integrity in the digital age. *MedEdPublish*, 8, Article 142. <https://doi.org/10.15694/mep.2019.000142.1>
- Yusuf, A., Pervin, N., & Román-González, M. (2024). Generative AI and the future of higher education: A threat to academic integrity or reformation? Evidence from multicultural perspectives. *International Journal of Educational Technology in Higher Education*, 21(1), Article 21. <https://doi.org/10.1186/s41239-024-00453-6>
- Zhang, G., Zhang, T., Mao, S., Xu, Q., & Ma, X. (2023). Supervisors' ethical leadership and graduate students' attitudes toward academic misconduct. *PLoS One*. <https://doi.org/10.1371/journal.pone.0283032>
- Zhang, X. (2025). Research misconduct in Chinese universities: Towards an emotional analysis. *Journal of Academic Ethics*, 23(2), 1953–1970. <https://doi.org/10.1007/s10805-025-09636-w>
- Zhao, L., Peng, J., Yang, X., Yan, W., Ke, S., Batool, K., & Lee, K. (2024). Effects of academic dishonesty policy reminder on university students' exam cheating—a double-blind randomized controlled experimental field study. *Studies in Higher Education*, 49(4), 592–608. <https://doi.org/10.1080/03075079.2023.2246073>
- Zia, A., & Khan, M. M. (2023). Addressing contract cheating in Pakistani higher education: Strategies for upholding academic integrity. *Journal of Professional Research in Social Sciences*, 10(2), 84–98. <https://doi.org/10.58932/MULA0015>

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